MINTS: Testing Strict Motif-Detector Claims in Nucleotide Transformers

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Abstract

Genomic foundation models often make biological labels linearly accessible, but decodability alone does not identify an internal mechanism. We study whether attention heads in nucleotide transformers can be justified as biological motif detectors under a strict mechanistic evidence standard. MINTS applies tokenization-aware QK/OV circuit analysis, residual-stream probing with controls, matched attention enrichment, activation patching, and sparse feature search to DNABERT-2 and a tested Nucleotide Transformer backend. Before making mechanistic claims, we report raw-sequence task baselines: GC-only AUROC ranges from 0.6361 to 0.9088, and 3–6-mer TF-IDF AUROC ranges from 0.7956 to 0.9406. DNABERT-2 layer-11 residual readouts achieve AUROC values of 0.9137, 0.9383, 0.8954, and 0.8847 on promoter and splice-site tasks, showing strong label decodability rather than a circuit-level explanation. Controls sharpen this interpretation: random-label and position-only probes are near chance; splice-site probes remain about 0.25 AUROC above GC-only controls; promoter probes are close to strong GC/k-mer baselines and should be interpreted cautiously. The primary strict test is negative: across 51,249 GM12878 CTCF sequences, no DNABERT-2 head meets the registered QK-to-motif criterion or matched attention-enrichment criterion. Batch activation patching finds promoter-TATA over-restoration and weaker splice-donor restoration, but these task-specific causal signals do not establish a CTCF motif-detector circuit. The results provide a calibrated evidence standard for nucleotide-transformer interpretability and show where representation-level success falls short of a strict motif-detector claim.

1 Introduction

Transformer models for nucleotide sequences now perform well on regulatory and splice-site prediction tasks [8, 4, 7]. However, much of the usual evidence for biological understanding remains predictive or post hoc: attribution maps, motif visualizations, or probes can identify useful correlations, but they do not by themselves explain how internal computations produce predictions [1, 12]. This distinction is central for mechanistic interpretability, where a claim about a component should survive direct tests of what it reads, where it attends, what it writes, and whether intervention on it changes the relevant behavior [5, 10, 6].

Nucleotide transformers are a useful setting for this question. Known motifs give natural ground-truth hypotheses: a motif can specify where a component would need to attend, what feature it would need to read, and which perturbation should matter. They also expose a common interpretability risk. Attention maps, motif visualizations, and high probe scores can suggest a biological mechanism even when the model is using GC content, genomic-coordinate artifacts, distributed features, or another correlated signal. MINTS is designed to make the required evidence explicit.

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The motifs used here are intentionally canonical. CTCF is a DNA-binding factor with a well-studied binding motif and public ENCODE peak calls in GM12878 cells, making it a natural strict test case for motif-local attention. The TATA box is a short promoter-associated sequence whose mutation gives a simple promoter perturbation. Splice donor and acceptor dinucleotides mark exon-intron boundaries and give compact sequence edits for task-level causal tests. These examples are not meant to exhaust regulatory biology; they are selected because a correct circuit-level claim should specify nucleotide support, token support, and an intervention target.

This paper contributes a calibrated evidence standard and a completed case study. MINTS separates three claims that are often conflated: residual decodability, task-level causal restoration under activation patching, and strict circuit-level motif detection. The primary strict test is CTCF: under the registered QK-alignment and matched-enrichment criteria, neither tested backend yields a CTCF attention head that should be called a strict motif detector. The promoter and splice analyses play a supporting role by showing that task labels can be decodable and that interventions can produce task-specific causal signals without establishing the stricter CTCF circuit claim.

2 Related Work and Positioning

Mechanistic interpretability of transformers has developed standard tools for decomposing attention heads, locating causal pathways, and tracing computational graphs [5, 10, 2]. QK/OV decomposition is especially relevant here because it separates where a head attends from what it writes. Activation patching then gives an intervention-based test of whether the component matters for a behavior [6]. MINTS applies these established tools to genomic transformers with explicit motif-level criteria, using ground-truth motifs to test both positive circuit evidence and possible overinterpretation.

Genomic deep-learning interpretability has a different tradition. Models such as DeepBind, SpliceAI, DNABERT, GPN, and Nucleotide Transformer show that neural sequence models learn useful biological regularities [1, 7, 8, 3, 4]. Many analyses in this area focus on motif visualization, attribution, saliency, or downstream probing. These are valuable, but they usually do not establish an internal algorithm. A saliency map may highlight a motif, and a probe may decode a label, while the model's actual computation may depend on a distributed feature, a correlated composition signal, or a different component entirely.

MINTS is positioned between these literatures. It does not claim to solve genomic interpretability in general, and it does not treat a high downstream or probe score as mechanistic evidence. Instead, it asks a narrower question: when, if ever, is it justified to call a specific attention head a biological motif detector? This choice deliberately favors falsifiable criteria over narrative completeness. Negative results are therefore useful evidence, because they identify where an appealing mechanistic story is not supported by the implemented tests.

3 Methods and Evidence Criteria

Biology primer. Table 1 defines the biological and tokenization terms used in the evidence stack.

Transformer circuits. Let $\tau(x) = (t_1, \ldots, t_T)$ tokenize a nucleotide sequence into vocabulary items. The initial hidden state is

$$H^{(0)}_{j,:} = E_{\text{tok}}[t_j, :] + P_{j,:}. \tag{1}$$

For head h in layer l, with hidden state $H^{(l-1)}$, the attention pattern is

$$A_h^{(l)} = \operatorname{softmax}\left(\frac{H^{(l-1)} W_h^Q (W_h^K)^T (H^{(l-1)})^T}{\sqrt{d_k}}\right), \tag{2}$$

and the head output is $A_h^{(l)} H^{(l-1)} W_h^V W_h^O$. Following transformer-circuit notation [5], the QK circuit $W_h^Q (W_h^K)^T$ controls attention scores, while the OV circuit $W_h^V W_h^O$ controls what is written to the residual stream. For nucleotide models, tokenization matters: a motif position may correspond to one nucleotide token, a fixed k-mer token, or a learned BPE span.

Table 1: Glossary for mechanistic-interpretability readers.

Term	Meaning in this paper
Motif	A recurring DNA pattern associated with a biological function, such as transcription-factor binding or splice recognition.
PWM	Position weight matrix: a per-position nucleotide model used to score how well a DNA window matches a motif.
JASPAR	A public database of curated transcription-factor binding motifs; MINTS uses the MA0139.1 CTCF matrix.
CTCF	CCCTC-binding factor, a DNA-binding protein involved in chromatin organization and regulatory insulation.
TATA box	A short A/T-rich promoter element used here for promoter perturbation and patching tests.
Promoter	A regulatory DNA region near a gene start site that can influence transcription initiation.
Splice donor/acceptor	Boundary signals at the start and end of introns; donor sites usually contain GT and acceptor sites usually contain AG in genomic DNA.
Nucleotide token	A model input unit corresponding to one or more DNA characters after tokenization.
k-mer	A fixed-length DNA substring of length k, such as a 6-mer token.
BPE	Byte-pair encoding, a learned variable-length tokenizer; DNABERT-2 BPE tokens can cover different numbers of nucleotides.

Table 2: Registered evidence criteria for candidate motif circuits. Thresholds are conservative screens: they are designed to prevent upgrading weak correlations, near-background attention, or unstable patching effects into motif-detector claims.

Claim	Statistic	Threshold	Purpose and scale
Motif localization	Attention enrichment ratio ρ_h	$\rho_h \geq 2.0$	Requires at least doubled motif-support attention over matched background.
Motif ranking	QK/motif Pearson correlation	$r \geq 0.5,\ p < 0.05$	Requires a large effect size, not merely significance from millions of tokens.
Causal importance	Patching metric $\mathrm{PM}(h, l)$	$\mathrm{PM} \geq 0.5$	Requires at least half restoration of the clean-minus-corrupted target effect.
Strong causal importance	Same	$\mathrm{PM} \geq 0.75$	Marks stronger recovery while leaving room below exact full restoration.
Residual decodability	Probe AUROC or macro-F_1	≥ 0.80 or $+0.10$	Establishes diagnostic readout quality, not circuit use by the model.
Robustness	Matched negative controls	No matched artifact	Requires simple controls not to explain the candidate signal.

Tokenization-aware motif support. The strict CTCF tests use nucleotide coordinates as ground truth and then map them to model-token coordinates. Let a motif hit span the half-open nucleotide interval $[a, b)$ and let token j span $[u_j, v_j)$. Token j supports the motif when

$$\max(0, \min(v_j, b) - \max(u_j, a)) \geq 1, \tag{3}$$

that is, when the token overlaps the motif interval by at least one nucleotide. Special tokens with zero-width offsets are retained for hidden-state alignment but receive no motif support. This rule is implemented in `src/motif_scoring.py` and recorded in token-score manifests as `token_support_min_overlap_bp=1`.

Sequence-to-token support mapping.

1. Score each nucleotide window with the JASPAR CTCF PWM and mark motif windows above the deterministic support threshold.
2. Convert each motif window to a half-open nucleotide interval $[a, b)$.
3. Tokenize the same sequence with offsets, retaining zero-width special tokens for alignment.
4. Mark every token whose nucleotide interval overlaps $[a, b)$ by at least one base.
5. Use the resulting token-support intervals for QK/motif correlation and matched attention-enrichment tests.

Strict motif-detector criterion. An attention head is treated as a candidate biological motif detector only when three signals converge. It must assign elevated attention mass to motif-support positions, its QK scores must rank motif-support tokens above matched background tokens, and patching its activation must restore the task-relevant scalar in motif-destroying counterfactuals. The registered thresholds used here are in Table 2. These thresholds intentionally make attention alone insufficient.

Activation patching. For a clean sequence x_c and motif-destroyed corrupted sequence x_{corr}, we patch a head output from the clean run into the corrupted run. The restoration metric is

$$\mathrm{PM}(h, l) = \frac{f(x_{\mathrm{patched}}) - f(x_{\mathrm{corr}})}{f(x_c) - f(x_{\mathrm{corr}})}, \tag{4}$$

where f is a scalar target such as a probe decision function or class logit. $\mathrm{PM} = 0$ indicates no recovery, $\mathrm{PM} = 1$ indicates full recovery of the clean effect, and values above one indicate over-restoration rather than simple full recovery [6].

Residual probes and controls. Linear probes are trained on frozen mean-pooled residual vectors. Their role is diagnostic: a high probe score establishes linear decodability, not that the probe direction is causal or used by the model. We therefore compare residual probes with GC-content-only classifiers, position-only classifiers where coordinates are available, GC-matched test subsets, shuffled-label probes, and GC-content distribution shifts. This framing follows linear-representation work while preserving the weaker interpretation that probes support [11].

4 Experimental Setup

The primary model is `zhihan1996/DNABERT-2-117M`, an encoder-style genomic transformer with learned BPE tokenization. The comparison backend is `InstaDeepAI/nucleotide-transformer-v2-100m-multi-species`, used as a tested fixed-6mer Nucleotide Transformer checkpoint [4]. The run exports DNABERT-2 QK and OV matrices for all 12 layers and 12 heads per layer, caches residual vectors for layers 0, 5, and 11, and uses custom Hugging Face forward hooks for attention-output patching because the installed TransformerLens path does not expose all required DNABERT-2 per-head tensors.

The residual-probe tasks are TATA promoter, non-TATA promoter, splice donor, and splice acceptor classification. The configured splits contain 5,062/212 train/test examples for TATA promoters, 30,000/1,372 for non-TATA promoters, and 30,000/3,000 for each splice task. The strict CTCF scan uses 51,249 prepared GM12878 CTCF peak sequences and the JASPAR MA0139.1 CTCF motif, aligned to exact overlapping DNABERT-2 BPE token spans. Batch patching uses motif-destroying clean/corrupted pairs for promoter-TATA and splice-donor tasks.

To avoid treating probes as end-to-end model performance, we report raw-sequence predictive baselines and a frozen DNABERT-2 sequence head before the mechanistic evidence. The raw-sequence baselines are a GC-content logistic classifier using GC fraction, GC skew, and sequence length, and a TF-IDF 3–6-mer logistic classifier. The frozen DNABERT-2 sequence head is a balanced logistic classifier trained on cached layer-11 sequence embeddings. Frozen residual readouts are shown beside these baselines only as representation context; neither DNABERT column is full encoder fine-tuning.

The pipeline writes small review-facing CSV, JSON, and figure artifacts while treating large activation, circuit-matrix, token-score, and sparse-autoencoder files as reproducible generated outputs. The full run takes about eight hours on the tested CUDA setup. The most expensive stages are the cross-model comparison and strict all-layer CTCF scans, so the appendix includes both full reproduction commands and a reduced smoke-run command for checking the code path on a smaller subset.

5 Results

Predictive performance context before mechanism. Table 3 separates task predictability from mechanistic evidence. The raw-sequence baselines are strong, especially on promoter tasks: a simple GC-content classifier already reaches 0.8955–0.9088 AUROC, and a 3–6-mer classifier reaches 0.9297–0.9406 AUROC. Splice tasks are less explained by composition alone, but k-mer baselines still recover substantial signal. The frozen DNABERT-2 layer-11 sequence head reaches 0.8847–0.9383 AUROC. The residual readout is included to show that the pretrained representation is predictive, but it remains a residual-decoding result rather than a circuit explanation.

Residual decodability is strong but not uniformly clean. DNABERT-2 layer-11 residual probes exceed the registered residual-decodability threshold for all four tasks. Table 4 combines headline AUROC values with the most important controls. Random-label probes collapse to chance, and position-only metadata does not explain the results. The splice-site tasks survive stronger checks: donor and acceptor residual probes remain about 0.25 AUROC above GC-only baselines on matched subsets and remain strong under GC shifts. The promoter tasks require caution because GC-only

Table 3: Predictive task context before mechanistic analysis. GC-only and 3–6-mer rows are raw-sequence baselines. The frozen DNABERT-2 sequence head is trained on cached layer-11 sequence embeddings; the residual readout is reported separately as diagnostic decodability evidence.

Task	Train	Test	GC AUROC	GC AUPRC	GC Acc.	k-mer AUROC	k-mer AUPRC	k-mer Acc.	Seq-head AUROC	Readout AUROC
promoter_tata	5,062	212	0.8955	0.9040	0.8019	0.9297	0.9378	0.8255	0.9137	0.9137
promoter_no_tata	30,000	1,372	0.9088	0.9218	0.8316	0.9406	0.9482	0.8652	0.9383	0.9383
splice_sites_donors	30,000	3,000	0.6560	0.6892	0.6013	0.8185	0.8093	0.7430	0.8954	0.8954
splice_sites_acceptors	30,000	3,000	0.6361	0.6343	0.5883	0.7956	0.7792	0.7220	0.8847	0.8847

Table 4: DNABERT-2 layer-11 probe-control summary. Numeric columns report AUROC; random-label values are means over eight shuffled-label runs.

Task	Resid.	GC-only	Pos.-only	GC-match	Rand.	GC-shift range
promoter_tata	0.9137	0.8956	0.4136	0.9137	0.4700	0.6182–0.8030
promoter_no_tata	0.9383	0.9088	0.3865	0.9383	0.5129	0.6312–0.8368
splice_sites_donors	0.8954	0.6560	0.4414	0.8944	0.5064	0.8432–0.8669
splice_sites_acceptors	0.8847	0.6361	0.4461	0.8838	0.5005	0.8573–0.8580

baselines are already high; their residual-probe advantage over GC-only controls is small on the matched comparison.

This distinction changes the interpretation of the headline probe table. The splice-site result is evidence that DNABERT-2's frozen residual stream contains linearly accessible information beyond simple GC composition or coordinate metadata. The promoter result is weaker mechanistically: it remains a valid decodability result, but the controls show that promoter-associated composition can explain much of the signal. This is why the paper avoids claiming that the promoter probes have isolated a clean causal promoter motif direction.

The strict CTCF motif-detector test fails. The CTCF scan produced 281,915 motif-support tokens and 1,843,874 finite token positions after BPE span correction. Across all 144 DNABERT-2 heads, the best QK-to-motif correlation is $r = 0.3004$, below the registered $r \geq 0.5$ criterion. The best matched attention-enrichment ratio is $\rho_h = 1.3130$, below the registered $\rho_h \geq 2.0$ criterion. The tested Nucleotide Transformer backend also fails to recover a strict CTCF candidate, with best $r = 0.0192$ and best $\rho_h = 1.00009$. Thus, the tested fixed-6mer backend does not support the CTCF motif-local head hypothesis under the same criteria.

The negative CTCF result is not a failure of the pipeline; it is the point of using a registered evidence stack. With more than 1.8 million finite token positions, small correlations can be statistically nonzero while remaining too small to support a mechanistic claim. The all-layer enrichment scan similarly prevents a common attention-overinterpretation mistake: a head can have visible structure without assigning enough mass to matched motif-support positions to count as a motif-local detector.

Threshold sensitivity supports the conservative conclusion. Table 6 and Figure 2 summarize threshold sweeps over QK correlation, matched enrichment, and auxiliary patching metrics. Relaxing the QK threshold alone yields 31 heads at $r \geq 0.1$, five at $r \geq 0.2$, one at $r \geq 0.3$, and none at $r \geq 0.4$. Relaxing matched enrichment alone yields three heads at $\rho_h \geq 1.1$, one at $\rho_h \geq 1.25$, and none at $\rho_h \geq 1.5$. The strict CTCF chain remains negative: no head jointly passes QK and enrichment once $r \geq 0.2$, even with $\rho_h \geq 1.1$, and the most permissive joint count at $r \geq 0.1, \rho_h \geq 1.1$ is two heads, equal to the 95th percentile of a permuted-head alignment null. Token-level motif-score nulls provide an additional calibration: the observed support-token count is 256,918, compared with a shuffled-score 95th percentile of 36,093.05 and zero nearest-GC non-support tokens above the motif-support threshold. The promoter and splice patching thresholds are more permissive because those are auxiliary task-specific interventions, not CTCF motif-detector proof.

Auxiliary promoter/splice patching gives task-specific causal signals. Batch promoter-TATA patching over 327 usable pairs identifies layer 4 head 8 with mean $\mathrm{PM} = 1.4029$; layer 6 head 8 and layer 2 head 7 are also high. Because $\mathrm{PM} > 1$ overshoots the clean-minus-corrupted effect, this is a methodologically sensitive causal signal rather than a simple full-recovery story. Batch splice-donor patching over 500 pairs identifies layer 1 head 8 with mean $\mathrm{PM} = 0.5485$, a weaker but threshold-crossing effect. These interventions support task-relevant internal computation, but they

Table 5: Target-alignment ledger. CTCF is the primary strict motif-detector chain. Promoter and splice results are auxiliary evidence for task-specific residual decodability and causal signal, not substitutes for a CTCF proof.

Target	Performance	Probe	QK	Enrichment	Patching	OV	SAE	Role
CTCF	51,249-sequence strict scan; no classifier target	No CTCF residual classifier used as strict evidence	Best $r = 0.3004$; fail	Best $\rho_h = 1.3130$; fail	No candidate promoted after failed QK/enrichment	No aligned OV claim	Best cosine 0.1158; weak	Primary strict chain
Promoter-TATA	GC 0.8955; k-mer 0.9297; seq-head/readout 0.9137 AUROC	Decodable, but close to strong baselines	Not target-aligned in current artifact	Not target-aligned	Best $\mathrm{PM} = 1.4029$; over-restores	L2H7 output cosine 0.1261; weak	Not promoter-specific	Auxiliary causal evidence
Splice donor	GC 0.6560; k-mer 0.8185; seq-head/readout 0.8954 AUROC	Decodable beyond GC and k-mer	Not target-aligned in current artifact	Not target-aligned	Best $\mathrm{PM} = 0.5485$; partial	Not audited	Not splice-specific	Auxiliary causal evidence

Table 6: Threshold sensitivity summary. Counts are heads passing each single-metric threshold, except the joint row, which requires both CTCF QK and enrichment. Null references include permuted head alignment plus token-level motif-score and GC-matched background calibration.

Target	Sweep	Permissive count	Registered/strict count	Best value	Null reference
CTCF	QK r	31 at $r \geq 0.1$	0 at $r \geq 0.5$	0.3004	Motif-score shuffle: 256,918 observed vs 36,093.05 p95
CTCF	Enrichment ρ_h	3 at $\rho_h \geq 1.1$	0 at $\rho_h \geq 2.0$	1.3130	Position-matched background mean $\rho_h \approx 1.00$
CTCF	Joint QK/enrichment	2 at $r \geq 0.1, \rho_h \geq 1.1$	0 at $r \geq 0.5, \rho_h \geq 2.0$	–	Permuted-head 95th percentile is 2 at the permissive setting
Promoter-TATA	Patching PM	40 at $\mathrm{PM} \geq 0.25$	16 at $\mathrm{PM} \geq 0.75$	1.4029	Auxiliary task-specific signal
Splice donor	Patching PM	3 at $\mathrm{PM} \geq 0.25$	0 at $\mathrm{PM} \geq 0.75$	0.5485	Auxiliary task-specific signal

do not identify a strict CTCF motif detector because the CTCF QK and enrichment screens have no passing head.

OV, SAE, and cross-model audits remain conservative. The TATA OV readout audit for layer 2 head 7 finds only weak alignment with the trained TATA probe direction: the top output-write singular-vector cosine is 0.1261 and the probe self-gain through the OV matrix is -0.0326. Sparse-autoencoder CTCF feature search is also weak, with best residual and MLP motif cosines of 0.0884 and 0.1158. In the protocol-level cross-model comparison, DNABERT-2 residual probes outperform the tested Nucleotide Transformer backend on the same tasks, but this is a checkpoint- and protocol-specific result, not a general claim about tokenization families or all Nucleotide Transformer variants.

6 What This Teaches Mechanistic Interpretability

The main lesson is that representation-level success and component-level mechanism are different claims. A residual probe can reveal that a biological label is linearly accessible while leaving open whether the model uses that direction, whether the feature is distributed, or whether the signal is driven by composition. Likewise, attention and motif visualizations can suggest a biological story without showing that a head's QK circuit ranks motif-support tokens or that intervention on the head changes the relevant behavior.

MINTS treats biological motifs as useful test cases precisely because they provide external ground truth. A motif defines nucleotide support, gives a concrete perturbation target, and lets us ask whether model-token support agrees with circuit behavior. The negative CTCF result is therefore informative rather than merely null: it shows that strict evidence stacks can prevent a plausible but unsupported motif-detector story. For mechanistic interpretability, the practical recommendation is to report claims at the level they are actually supported: decodability, task-specific causal signal, motif-local attention, or strict circuit-level motif detection.

7 Limitations

This study is intentionally scoped to small and medium genomic transformers and does not claim exhaustive explanation. Current large-model circuit-tracing work also emphasizes partial recovery of computation rather than complete mechanistic accounts [2, 9]. The strict CTCF failure applies to

Table 7: Evidence ledger. Each row states the claim, the required evidence standard, the observed result, the supported interpretation, and the main limitation.

Claim	Required evidence	Observed result	Interpretation	Limitation
Promoter and splice labels are decodable from DNABERT-2 residual states	Strong residual readout plus controls against trivial artifacts	Layer-11 AUROC range 0.8847–0.9383; random-label and position controls near chance	Residual states contain task-relevant information	Promoter tasks remain close to strong GC and k-mer baselines
Splice residual decodability is not explained by GC alone	Matched and shifted GC controls should preserve the readout advantage	Splice probes remain about 0.25 AUROC above GC-only matched controls	Stronger evidence for non-composition residual signal on splice tasks	Still a probe result, not a localized circuit
Promoter-TATA patching identifies task-specific causal signal	Head-level patching should restore a corrupted scalar target across many pairs	Best mean $\mathrm{PM} = 1.4029$ over 327 usable pairs	A head can causally amplify the promoter-TATA readout under patching	Over-restoration and probe geometry prevent a simple full-recovery claim
Splice-donor patching identifies a weaker causal signal	Patching should restore a corrupted splice-donor scalar target	Best mean $\mathrm{PM} = 0.5485$ over 500 usable pairs	Partial task-specific causal signal exists	Effect is small on average and not tied to QK/enrichment motif evidence
DNABERT-2 has a strict CTCF motif-detector head	One head should pass QK/motif ranking, motif-local enrichment, and causal evidence	Best QK $r = 0.3004$; best $\rho_h = 1.3130$; no passing candidate	The strict single-head CTCF detector claim is not supported	Negative result is scoped to selected model, motif scorer, data, and thresholds
Relaxed CTCF thresholds do not rescue the strict claim	Sensitivity sweep should reveal whether conclusions depend on the registered cutoffs	No joint QK/enrichment pass once $r \geq 0.2$, even at $\rho_h \geq 1.1$	The negative CTCF conclusion is not an artifact of one conservative cutoff	Per-head shuffled QK-score nulls require raw per-token QK vectors
SAE and OV audits do not reveal a simple alternate CTCF circuit	Output-write or sparse features should align strongly with the biological target	TATA OV cosine 0.1261; CTCF SAE best cosine 0.1158	Auxiliary audits remain consistent with a distributed or absent simple motif circuit	Audits are not exhaustive circuit discovery

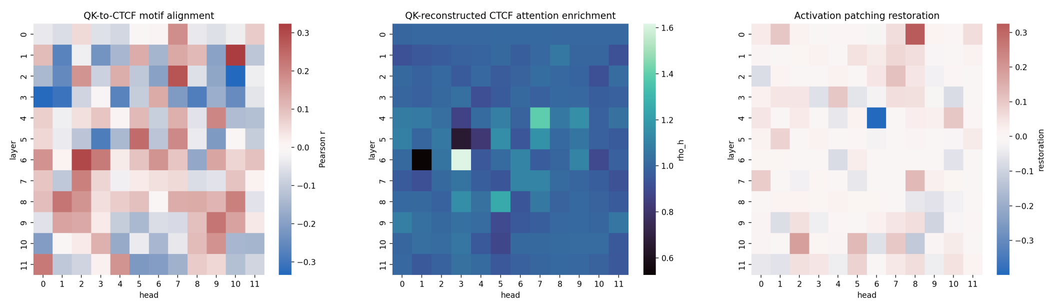

Figure 1: Mechanistic evidence heatmaps. Left: DNABERT-2 CTCF QK-to-motif correlations. Middle: matched attention-enrichment ratios. Right: promoter-TATA batch patching. The CTCF heatmaps show a negative strict-proof result, while the TATA patching heatmap shows task-specific causal over-restoration.

the selected checkpoints, GM12878 CTCF intervals, JASPAR MA0139.1 motif scorer, BPE span-alignment rule, selected causal targets, and registered thresholds. It is not an impossibility theorem for all genomic transformers, all CTCF representations, all cell types, all motif scorers, or all circuit decompositions.

Linear probes remain limited evidence. They can reveal decodable information while exploiting correlated sequence composition, split artifacts, or metadata. The added GC, position, matched, random-label, and shift controls reduce this risk, especially for splice-site tasks, but promoter decodability remains entangled with GC composition. Activation patching is causal but incomplete: head-level restoration does not characterize the full circuit, and over-restoration can reflect denominator sensitivity, probe geometry, or out-of-distribution patched states. The threshold-sensitivity table reduces concern that the CTCF conclusion depends on one arbitrary cutoff, but it does not replace new data-generating null experiments. Finally, biological annotations are incomplete, and a single JASPAR PWM cannot represent all relevant regulatory mechanisms.

8 Conclusion

MINTS turns a common interpretability temptation into a falsifiable test: before calling an attention head a biological motif detector, require aligned QK scores, motif-local attention enrichment, and causal restoration. On DNABERT-2, promoter and splice-site labels are strongly linearly decodable from late residual states, and splice-site decodability survives simple composition and position controls. But the strict CTCF motif-detector proof fails across all tested heads, and the task-specific

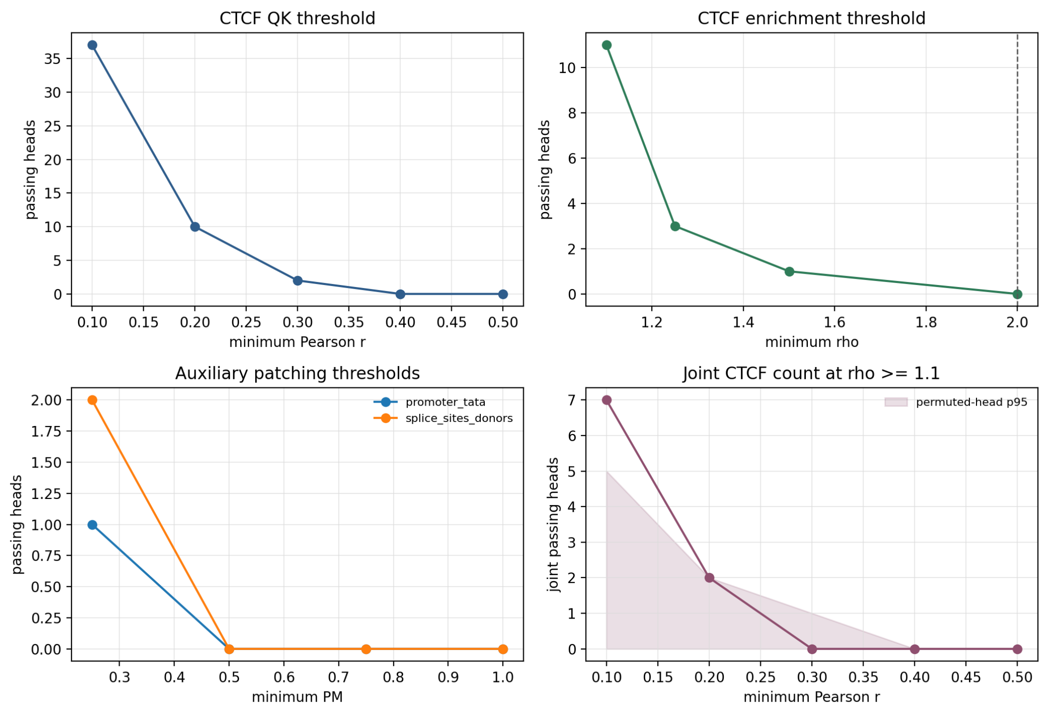

Figure 2: Threshold sensitivity for QK alignment, matched attention enrichment, auxiliary patching, and joint CTCF QK/enrichment counts. Individual relaxed screens admit a small number of heads, but the strict CTCF chain remains negative under registered thresholds and under joint relaxation once $r \geq 0.2$.

patching effects do not change that conclusion. The contribution is therefore both positive and negative: a reproducible evidence standard for nucleotide-transformer interpretability, and a concrete demonstration that strong representation-level decodability should not be upgraded into a circuit-level biological claim without stronger evidence.

References

[1] Babak Alipanahi, Andrew Delong, Matthew T. Weirauch, and Brendan J. Frey. Predicting the sequence specificities of DNA- and RNA-binding proteins by deep learning. *Nature Biotechnology*, 33(8):831–838, 2015. doi: 10.1038/nbt.3300.

[2] Emmanuel Ameisen, Jack Lindsey, Adam Pearce, Wes Gurnee, Nicholas L. Turner, Brian Chen, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. Circuit tracing: Revealing computational graphs in language models. *Transformer Circuits Thread*, 2025. URL `https://transformer-circuits.pub/2025/attribution-graphs/methods.html`.

[3] Gonzalo Benegas, Sanjit Singh Batra, and Yun S. Song. DNA language models are powerful predictors of genome-wide variant effects. *Proceedings of the National Academy of Sciences*, 120(44):e2311219120, 2023. doi: 10.1073/pnas.2311219120.

[4] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P. de Almeida, Hassan Sirelkhatim, Guillaume Richard, Marcin Skwark, Karim Beguir, Marie Lopez, and Thomas Pierrot. Nucleotide transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22(2):287–297, 2025. doi: 10.1038/s41592-024-02523-z.

[5] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. https://transformer-circuits.pub/2021/framework/index.html.

[6] Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024. doi: 10.48550/arXiv.2404.15255.

[7] Kishore Jaganathan, Sofia Kyriazopoulou Panagiotopoulou, Jeremy F. McRae, Siavash Fazel Darbandi, David Knowles, Yang I. Li, Jack A. Kosmicki, Juan Arbelaez, Wenwu Cui, Grace B. Schwartz, Eric D. Chow, Efstathios Kanterakis, Hong Gao, Amirali Kia, Serafim Batzoglou, Stephan J. Sanders, and Kyle Kai-How Farh. Predicting splicing from primary sequence with deep learning. *Cell*, 176(3):535–548.e24, 2019. doi: 10.1016/j.cell.2018.12.015.

[8] Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V. Davuluri. DNABERT: pre-trained bidirectional encoder representations from transformers model for DNA-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021. doi: 10.1093/bioinformatics/btab083.

[9] Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the biology of a large language model. *Transformer Circuits Thread*, 2025. URL `https://transformer-circuits.pub/2025/attribution-graphs/biology.html`.

[10] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2202.05262.

[11] Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 39643–39666, 2024. URL `https://proceedings.mlr.press/v235/park24c.html`.

[12] Haopeng Yu, Heng Yang, Wenqing Sun, Zongyun Yan, Xiaofei Yang, Huakun Zhang, Yiliang Ding, and Ke Li. An interpretable RNA foundation model for exploring functional RNA motifs in plants. *Nature Machine Intelligence*, 6:1616–1625, 2024. doi: 10.1038/s42256-024-00946-z.

A Theoretical Details

Definition 1 (Biological Motif Detector)**.** *An attention head h in layer l functions as a biological motif detector for motif m if there exists a nonempty set of relevant query positions $\mathcal{Q}_m$ and a threshold τ such that, for each $i \in \mathcal{Q}_m$,*

$$\mathbb{E}\left[\sum_{j \in \mathrm{supp}_m(x)} A_h^{(l)}[i,j] \mid \mathrm{supp}_m(x) \neq \varnothing\right] \geq \tau \tag{5}$$

and the corresponding expectation conditioned on $\mathrm{supp}_m(x) = \varnothing$ is below τ. Here $\mathrm{supp}_m(x)$ is the set of key positions whose local sequence windows instantiate motif m.

Theorem 1 (Layer-1 Motif Detection via QK Circuits)**.** *Consider a layer-1 attention head operating on $H^{(0)}$. Let $s_m(j)$ denote a scalar motif score associated with key position j. If for a fixed query position i there are constants $a_i \in \mathbb{R}$ and $\gamma > 0$ such that*

$$q_i^T k_j = a_i + \gamma s_m(j) \tag{6}$$

for all admissible key positions j, then the induced attention distribution is monotonically increasing in $s_m(j)$. If motif-bearing positions are separated from background positions by a positive margin in s_m, the head is a motif detector for m.

Proof. The attention logit is $\ell_{ij} = q_i^T k_j/\sqrt{d_k} = a_i/\sqrt{d_k} + (\gamma/\sqrt{d_k})s_m(j)$. Since $\gamma/\sqrt{d_k} > 0$, scaling preserves the ordering of motif scores. The exponential and softmax are strictly monotone in each logit when other logits are fixed, so positions with larger motif scores receive larger attention weights. A positive separation margin therefore induces an attention-mass threshold that separates motif-bearing from background positions. □

Lemma 1 (Approximate Extension Under Linear Decodability)**.** *Suppose that at layer $l-1$ there exists a vector u_m such that*

$$u_m^T h_j^{(l-1)} = s_m(j) + \epsilon_j, \qquad |\epsilon_j| \leq \delta. \tag{7}$$

If a head's QK score is affine in this decodable direction and the clean motif/background score margin is greater than 2δ, then the attention ordering between motif and background positions is preserved.

Proof. For motif-bearing j_m and background j_b, the score difference is proportional to $s_m(j_m) - s_m(j_b) + \epsilon_{j_m} - \epsilon_{j_b}$. The perturbation term lies in $[-2\delta, 2\delta]$. A clean separation greater than 2δ keeps the sign of the score difference fixed, so softmax preserves the motif/background ordering. □

B Implementation and Reproducibility Details

The completed run started at 13:27:07 local time on April 14, 2026 and ended at 21:26:30 local time the same day. The manifest reports 28,760.213 seconds, or 7.989 hours. The longest steps were cross-model tokenization comparison, strict mechanistic proofs, and systematic causal intervention. The inspected result tree contains 125 files totaling about 4.71 GB, including CSV/JSON summaries, figures, NPZ archives, and sparse-autoencoder checkpoints.

The anonymized project repository is available at `https://anonymous.4open.science/r/MINTS/`. It contains the project code, configuration, local paper source, and reproduction instructions. The repository currently serves as the single source of truth for the submitted code; an explicit repository license has not yet been added.

The main reproduction commands are:

```
git clone https://anonymous.4open.science/r/MINTS/ MINTS
cd MINTS
python -m pip install -r requirements.txt
python main.py
python main.py --only-probe-controls
```

A smaller smoke run is:

```
python main.py --max-probe-train 512 --max-probe-test 256
  --max-qk-alignment-sequences 128
  --max-patching-pairs 32
  --max-cross-model-qk-alignment-sequences 128
  --max-feature-search-sequences 128 --sae-epochs 1
```

Important verification artifacts include:

- `results/pipeline_run.json`
- `results/tables/linear_probe_metrics.csv`
- `results/tables/linear_probe_controls.csv`
- `results/tables/downstream_task_performance.csv`
- `results/tables/target_alignment_table.csv`
- `results/tables/review_issue_matrix.csv`
- `results/tables/evidence_ledger.csv`
- `results/tables/threshold_sensitivity.csv`
- `results/qk_alignment/ctcf_qk_alignment.csv`
- `results/enrichment/ctcf_qk_alignment_matched_attention_enrichment.csv`
- `results/patching/promoter_tata_batch_dnabert_activation_patching.csv`
- `results/patching/splice_sites_donors_batch_dnabert_activation_patching.csv`

The heatmaps used in the paper are generated under `results/figures/`.

C Additional Results

The full configured-split residual probe AUROCs are 0.9137 for TATA promoters, 0.9383 for non-TATA promoters, 0.8954 for splice donors, and 0.8847 for splice acceptors. The corresponding AUPRC values are 0.9241, 0.9475, 0.9049, and 0.8954.

D Response-to-Review Checklist

Table 8 records the review issue matrix used to guide the revision. The anonymized reviewer IDs are treated as issue sources rather than as claims about reviewer intent.

Table 8: Review issue matrix.

ID	Issue	Concern	Resolution	Status
mV6D	Clarity	Paper read as several analyses.	Abstract/introduction now state one calibrated evidence-standard claim; Results opens with task context and a target ledger.	Addressed
mV6D	Target mismatch	Promoter/splice evidence blurred into CTCF claims.	Promoter/splice patching is auxiliary; CTCF remains the strict chain.	Addressed
C5jL	Missing performance	Mechanistic claims lacked predictive context.	Added GC and 3–6-mer baselines, a frozen DNABERT sequence head, and frozen-readout context before mechanism.	Addressed
C5jL	MI contribution	Venue contribution needed sharpening.	Contribution is now a falsifiable standard for motif-detector claims.	Addressed
C5jL	Thresholds	Registered thresholds need scale/sensitivity.	Added threshold purposes, sweeps, and available null calibration.	Addressed
ndFG	Undefined biology	Motifs and biological terms need more support.	Added motif rationale, glossary, and nucleotide-to-token support rule.	Addressed
ndFG	Limited scope	Conclusions could overstate selected models/motifs.	Limitations and target ledger state the scoped negative CTCF result.	Addressed

For promoter-TATA batch patching, 327 token-shape-preserving pairs were usable with no near-zero denominator failures. The best mean score was layer 4 head 8 with $\mathrm{PM} = 1.4029$, followed by layer 6 head 8 with $\mathrm{PM} = 1.3791$ and layer 2 head 7 with $\mathrm{PM} = 1.3642$. For splice donors, 500 pairs

Table 9: Response-to-review checklist with exact fixes.

Concern	Exact fix	Where it appears	Artifact
Writing clarity	Reframed the paper around one calibrated evidence-standard claim and one primary negative CTCF case study	Abstract, Introduction, target-alignment ledger, evidence ledger	`results/tables/evidence_ledger.csv`
Target mismatch	Separated CTCF strict motif-detector evidence from promoter/splice auxiliary decodability and patching evidence	Target-alignment table and evidence ledger	`results/tables/target_alignment_table.csv`
Missing performance context	Added GC and 3–6-mer raw-sequence baselines plus a frozen DNABERT sequence head before mechanistic claims, while keeping frozen readouts labeled as residual decodability	Predictive performance context subsection	`results/tables/downstream_task_performance.csv`
Weak MI contribution	Stated the contribution as a falsifiable standard for motif-detector claims rather than broad genomic interpretability	Introduction, Related Work, and "What This Teaches Mechanistic Interpretability"	Manuscript text
Threshold justification	Added threshold purposes, sensitivity sweeps, and available null calibration	Criteria table and threshold-sensitivity subsection	`results/tables/threshold_sensitivity.csv`
Undefined biology	Added motif rationale, glossary, and nucleotide-to-token overlap rule	Introduction and Methods glossary/mapping block	`src/motif_scoring.py`
Limited scope	Strengthened limitations to selected checkpoints, intervals, scorer, tokenization rule, causal targets, and thresholds; removed impossibility framing	Limitations and Conclusion	Manuscript text

were usable with no denominator failures, and the strongest mean restoration was layer 1 head 8 with $\mathrm{PM} = 0.5485$.

For the tested Nucleotide Transformer backend, residual-probe AUROCs were 0.6502, 0.6976, 0.5695, and 0.5647 on the same four tasks. DNABERT-2 therefore outperformed this backend by 0.2408–0.3259 AUROC in this protocol. This comparison should not be read as a general ranking of tokenizers or genomic transformer families.

E Declaration of LLM Usage

LLM-assisted tools were used during the project for mathematical exposition refinement, debugging support, and code-generation drafts through Codex with state-of-the-art GPT-5.4/GPT-5.5-class models. The research direction, core problem decomposition, step-by-step implementation decisions, and final scientific interpretation were directed by the author. The author reviewed the generated code, checked it for errors and failure modes, corrected mistakes, and takes responsibility for the final manuscript, code, and claims.

NeurIPS Paper Checklist

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